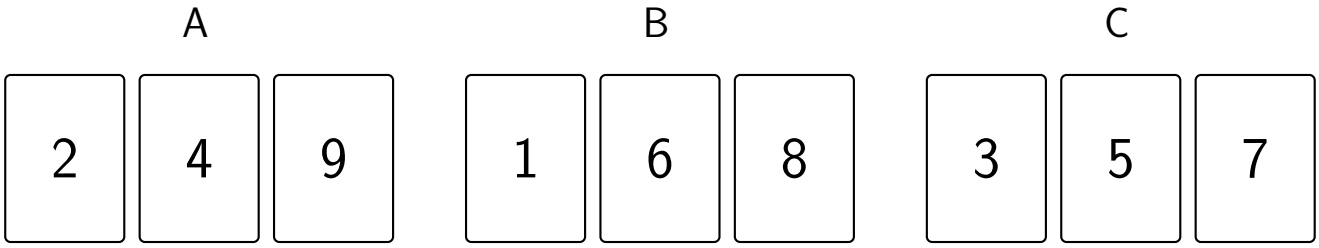
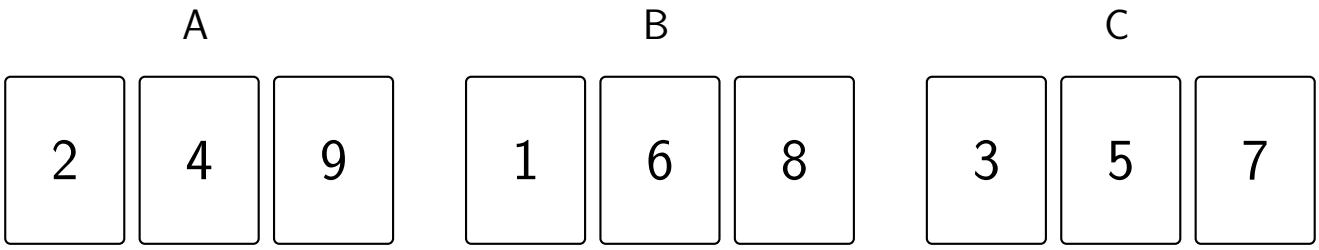
Each card has the same chance. Mix the bags separately. Draw once from each bag, then put both cards back. The bigger number wins.

Problem 1: One player picks one of the decks A, B, and C. Then the other player picks a different deck. They play ten rounds while a third player draws the cards and keeps score. Rotate jobs and play again. Is it better to pick first or second?



First pick	Second pick

Problem 2: For each pair of decks, find how many of the possible card pairs each deck wins. Is there a deck that wins more than half its pairs against both others? Do the three deck totals settle this question?



Problem 3: Keep 9 in A, 8 in B, and 7 in C. Use cards 1 to 6 once each to finish the decks so A wins more than half its pairs against B, B against C, and C against A. Find every arrangement and explain why none are missing.

